

Response to Reviewers

Dear Editor,

We sincerely thank the editor and reviewers for their careful reading of our manuscript and for the constructive comments. The comments helped us identify several places where the original manuscript did not sufficiently explain the data-generation workflow, the sampling strategy, the DDPG formulation, the optimizer output contracts, the physical checking procedure, and the climate boundary of the study. We have therefore revised the manuscript substantially rather than making only local edits.

The revised manuscript now explicitly distinguishes generated feasible block morphologies, morphology descriptors, analytic response generation, DNN surrogate prediction, descriptor-space optimization, nearest-feasible-block projection, physics-based cross-model stress testing, and cross-climate sensitivity analysis. We also reorganized the Supplementary Information to provide the response-generation equations, descriptor-dependency checks, surrogate validation results, optimizer-comparison details, projection analysis, physical stress-test outputs, and climate-sensitivity tables.

The main revisions are summarized in Table R1. We then provide point-by-point responses to all reviewer comments. In each response, we first address the reviewer's concern, then summarize the corresponding manuscript and Supplementary Information revisions, and finally state the interpretation boundary where appropriate.

Manuscript title: Surrogate-conditioned optimizer ranking in block-scale urban energy design: A comparison of policy learning and Pareto search

Manuscript ID: APEN-D-26-08148

Sincerely,
The Authors

CONTENTS

► Summary of major revisions	1
► Reviewer 1	4
■ Comment R1-1.	4
■ Comment R1-2.	4
■ Comment R1-3.	5

■	Comment R1-4.	5
■	Comment R1-5.	6
■	Comment R1-6.	7
■	Comment R1-7.	7
■	Comment R1-8.	8
■	Comment R1-9.	8
■	Comment R1-10.	9
■	Comment R1-11.	9
■	Comment R1-12.	10
■	Comment R1-13.	10
■	Comment R1-14.	11
■	Comment R1-15.	12
►	Reviewer 2.	13
■	Comment R2-1.	13
■	Comment R2-2.	13
■	Comment R2-3.	14
■	Comment R2-4.	15
■	Comment R2-5.	15
■	Comment R2-6.	16
■	Comment R2-7.	16
■	Comment R2-8.	17
►	Reviewer 3.	18
■	Comment R3-1.	18
■	Comment R3-2.	18
■	Comment R3-3.	19
■	Comment R3-4.	19
■	Comment R3-5.	20
►	Reviewer 4.	21
■	Comment R4-1.	21
■	Comment R4-2.	21
■	Comment R4-3.	22
■	Comment R4-4.	22
■	Comment R4-5.	23
►	Closing.	23

Summary of major revisions

The revised package reframes the study as a surrogate-conditioned, representation-aware optimizer comparison for early-stage urban energy design. The revision is not a light copy-editing pass. It rewrites the manuscript around the distinction between generated feasible morphology, analytic target assignment, surrogate prediction, optimizer retained outputs, feasible-block projection, and selected sensitivity analyses. This structure is now reflected in the Abstract, Introduction, Methodology, Results, Discussion, Conclusions, revised figures, revised tables, and Supplementary Information.

Table R1: Summary of major revisions made in response to the reviewers.

Revision package	Reviewer concern addressed	Main manuscript changes	Supplementary Information added	Key outcome
Manuscript reframing and abstract rewrite	Significance, readability, and scope	Rewrote the Abstract, Introduction, Contributions, Discussion, and Conclusions around surrogate-conditioned optimizer ranking	Added structured SI sections S1–S6 to support the revised narrative	The manuscript now presents a bounded design-support workflow, not a broad optimizer-superiority claim
Analytic response-generation clarification	Data source and objective definitions	Clarified that the 2000-sample dataset is an analytic response-generation dataset, not direct annual physical simulation for all samples	Added full EUI _t , EG, and H equations, noise terms, and clipping ranges in SI S1.4	The benchmark target source is explicit and reproducible
Sampling and descriptor-space coverage	Sample-generation strategy and 12 inputs	Rewrote morphology-generation and descriptor-construction sections; clarified that inputs are descriptors, not independent variables	Added prototype settings, descriptor dependencies, PCA coverage, OS _{LI} counts, and nearest-neighbour distance summaries in SI S1	Descriptor-space coverage is documented while geometric dependencies remain visible
Surrogate robustness assessment	Cross-validation and model reliability	Added repeated validation and grouped/boundary holdout discussion	Added repeated 5 x 5 CV, leave-one-OS _{LI} -out, outer-shell, feature-tail, residual, and scale-study results in SI S2	The selected surrogate is assessed within the analytic response-generation regime
DDPG formulation and reward correction	Questionable RL/MDP framing and reward equation	Rewrote DDPG as serialized static black-box search and corrected reward equations	Added seed-level DDPG behavior and retained-output explanations in SI S3	The RL terminology no longer implies physical-time morphology evolution
Optimizer output contracts and benchmark fairness	Output-contract asymmetry and optimizer ranking	Added equal-size comparison, output-contract table, and HV/IGD interpretation boundary	Added equal-size metrics, HV ceiling analysis, CMA-ES and RandomSearch details in SI S3	Descriptor-space ranking is tied to retained-output contracts

Table R1: Summary of major revisions made in response to the reviewers (continued).

Revision package	Reviewer concern addressed	Main manuscript changes	Supplementary Information added	Key outcome
Feasible-block projection analysis	Continuous descriptors versus feasible blocks	Added nearest-feasible-block projection and candidate-compression analysis	Added complete projection metrics and optimizer-linked gap decomposition in SI S5	Large descriptor archives are no longer equated with feasible-morphology diversity
Physics-based cross-model stress test	Insufficient or simplified physical checking	Added 24 locked cases with conservative interpretation	Added direct physical stress-test table and roof-irradiance unavailability note in SI S5	Weak metric and rank agreement are used as an interpretation boundary
Cross-climate sensitivity analysis	Specific climate-zone limitation	Added four-block sensitivity analysis for Beijing, Guangzhou, and Harbin	Added weather metadata, station-level changes, case-level climate results, and rank stability in SI S6	Climate effects are reported as limited sensitivity, not transfer proof
Code, data, figures, units, abbreviations, and readability	Code/data, abbreviations, units, figures, tables	Added abbreviations, corrected units, revised figures and captions, and updated Data and code availability	Added figure-data and result tables supporting manuscript and SI figures	The response package is more auditable and easier to read

The Supplementary Information was reorganized as an evidence store rather than as an appendix of loosely connected details. SI S1 now defines the generated block settings, morphology descriptors, descriptor dependencies, sampling coverage, and analytic response equations. SI S2 reports surrogate selection and robustness checks. SI S3 documents DDPG behavior and optimizer-output contracts. SI S4 separates morphology interpretation from optimizer scoring. SI S5 reports feasible projection and the physics-based cross-model stress test. SI S6 reports the limited cross-climate case analysis.

Table R2: New validation and sensitivity analyses added or clarified in the revision.

Analysis	Where added	Cases or folds	Main result	Interpretation boundary
Repeated 5 x 5 CV	Manuscript Table 3; SI S2.2	25 folds	nMAE = 0.0176, 0.0265, 0.0145 for EUIt, EG, and H	Surrogate accuracy within the analytic response-generation regime
Leave-one-OSLI-out	Manuscript Table 3; SI S2.4	9 held-out groups	nMAE = 0.0185, 0.0342, 0.0185	Grouped descriptor stress test, not an external physical validation
Outer-shell holdout	Manuscript Table 3; SI S2.5	boundary hold-out	nMAE = 0.0278, 0.0335, 0.0208	Boundary behavior is weaker than random-fold interpolation

Table R2: New validation and sensitivity analyses added or clarified in the revision (continued).

Analysis	Where added	Cases or folds	Main result	Interpretation boundary
Feature-tail holdout	Manuscript Table 3; SI S2.5	target-tail hold-out	nMAE = 0.0230, 0.0287, 0.0198	Tail behavior remains analytic-response based
Nearest-feasible-block projection	Manuscript Figure 8; SI S5.1	retained optimizer outputs	NSGA-II contracts from 2000 descriptor candidates to 51 unique feasible blocks	Descriptor archive size is not feasible-block diversity
Physics-based cross-model stress test	Manuscript Figure 9; Table 6; SI S5	24 locked cases	18 direct cases give EUIt nMAE 0.340, EG-proxy nMAE 0.405, H nMAE 0.601; Spearman values are -0.404, -0.023, and -0.043	Metric agreement is weak and ranking transfer is unsupported
Cross-climate sensitivity analysis	Manuscript Figure 10; Table 6; SI S6	4 blocks x 3 climates	EG Spearman rho = 1.0 in all three climates; H rho = 0.2, 0.4, 0.2; Harbin EUIt rho = 0.2	Limited selected-block sensitivity, not cross-climate generalization

These additions intentionally keep the interpretation boundary visible. The evidence supports a controlled, surrogate-conditioned optimizer comparison and a more transparent feasible-design interpretation. It does not establish general physical equivalence between the analytic response surface and Energy-Plus/Radiance outputs, and it does not show that optimizer rankings transfer unchanged across physical or climate settings.

► Reviewer 1.

✓ We thank Reviewer 1 for the careful reading and for the suggestions on readability, figures, units, software context, model assumptions, cross-validation, and broader audience accessibility. The revised manuscript responds by simplifying the framing, defining the main terms and symbols, expanding the method description, and keeping all claims within the analytic response-generation and limited sensitivity-analysis scope.

■ Comment R1-1.

The reviewer noted that the Abstract was too long and that the significance should be expressed more briefly and clearly.

Response.

✓ We thank the reviewer for this suggestion. We rewrote the Abstract to focus on the research significance without presenting the work as a direct claim of optimizer superiority. The revised Abstract now emphasizes surrogate-conditioned optimizer ranking, feasible-block projection, the physics-based cross-model stress test, and the cross-climate sensitivity analysis.

Revisions made in the manuscript and Supplementary Information.

The change appears in the revised Abstract and in the revised Introduction, where the study is framed as a design-support assessment of optimizer ranking under a shared surrogate response surface.

Relevant revised manuscript and supporting material.

Revised manuscript, Abstract.

The revised Abstract states that feasible block layouts are generated first, converted into 12 morphology descriptors, assigned EUI, EG, and H by a morphology-to-performance response surface, and then used to train the selected DNN surrogate. It reports repeated five-fold nMAE values of 0.0176, 0.0265, and 0.0145 for EUI, EG, and H, and it concludes that optimizer rankings should be interpreted together with projection and sensitivity-analysis boundaries.

■ Comment R1-2.

The reviewer noted that the manuscript contained too many acronyms and requested clearer symbols.

Response.

✓ We agree that the earlier version used too many abbreviations without enough support for readers outside the narrow optimization community. We added an Abbreviations block and clarified the symbols used for EUI, EG, H, DNN, DDPG, NSGA-II, HV, IGD, nMAE, and OSLL.

Revisions made in the manuscript and Supplementary Information.

The revised manuscript defines the main abbreviations in the front matter and introduces the morphology descriptors in Table 1 with symbols, units or type, construction notes, and interpretation.

Relevant revised manuscript and supporting material.

Revised manuscript, Abbreviations and Table 1.

The abbreviation list now defines the core energy, learning, optimization, and validation terms before the main text. Table 1 lists the 12 morphology descriptors used by the surrogate and optimizer interface and notes that these quantities are derived morphology descriptors rather than freely separable design inputs.

■ Comment R1-3.

The reviewer requested a more detailed caption for Figure 1.

Response.

✓ We revised the Figure 1 caption and the surrounding Methodology text so that the figure explains the full workflow rather than only showing a high-level schematic. The caption now identifies feasible morphology generation, analytic response assignment, surrogate-assisted optimization, feasible-block projection, and sensitivity analysis.

Revisions made in the manuscript and Supplementary Information.

The change appears in Figure 1 and Section 2. The revised caption avoids implying that all 2000 generated samples came from direct annual EnergyPlus and Radiance calculations.

Relevant revised manuscript and supporting material.

Revised manuscript, Figure 1 caption and Section 2.

Figure 1 is now described as a workflow linking feasible morphology generation, analytic response assignment, surrogate-assisted optimization, feasible-block projection, and sensitivity analysis. The Methodology text distinguishes generated feasible block geometries, 12 morphology descriptors, and selected locked block cases examined by physics-based and climate-sensitive calculations.

■ Comment R1-4.

The reviewer requested references for the Chinese context and the software platform, and asked why the selected tools are suitable for the analysis.

Response.

✓ We revised the Introduction to better connect the work to Chinese urban renewal, rooftop photovoltaic

deployment, and climate-resilient planning. We also clarified the software context by distinguishing the analytic response-generation dataset from the selected physics-based case calculations. This avoids suggesting that the full 2000-sample benchmark was produced by direct physical simulation.

Revisions made in the manuscript and Supplementary Information.

The changes appear in Section 1 and in the evaluation-mode description in Section 2.2. The Supplementary Information describes the analytic response generator and the case settings used for selected physics-based calculations.

Relevant revised manuscript and supporting material.

Revised manuscript, Sections 1.1 and 2.2.

The Introduction now states that Chinese urban districts require early-stage assessment of renewal, photovoltaic deployment, and climate-resilient planning before detailed building-system design is available. The Methodology now states that the analytic response generator assigns EUI_t, EG, and H to the generated morphology pool, while selected physics-based case calculations are conducted subsequently on locked generated blocks.

Supplementary Information, Section S1.

The Supplementary Information provides the case settings, morphology generation sequence, descriptor definitions, and analytic response equations used to construct the target-assignment surface.

■ Comment R1-5.

The reviewer asked why 12 parameters were selected for the building models.

Response.

✓ We clarified that the 12 quantities are morphology descriptors calculated after feasible block generation. They describe density, height variation, aspect-ratio conditions, sky-view openness, footprint density, open-space ratio, compactness, orientation, and open-space position. They are not presented as freely separable design inputs.

Revisions made in the manuscript and Supplementary Information.

The change appears in Section 2.1 and Table 1. The Supplementary Information adds dependency checks for the descriptor set.

Relevant revised manuscript and supporting material.

Revised manuscript, Section 2.1 and Table 1.

The revised text explains that the generator first constructs feasible block layouts and only then calculates descriptor values. Table 1 lists FAR, SD, AF, AR_{ew}, AR_{ns}, SVF, BD, OSR, SC, PAR, theta,

and OSLI with units or type and interpretation.

Supplementary Information, Section S1.3.

The descriptor dependency check reports that FAR is numerically linked to BD and AF, OSR is linked to footprint area and gross floor area, OSLI is integer-valued, six principal components explain 95% of descriptor variance, and exact duplicate descriptor rows are absent in the canonical 2000-row pool.

■ **Comment R1-6.**

The reviewer suggested adding an explanatory column to Table 1 and noted that Table 2 seemed to describe baseline or operating parameters rather than assumptions.

Response.

✓ We revised the tables to make their roles clearer. Table 1 now describes morphology descriptors with construction and interpretation notes. Table 2 has been reframed as an evaluation-mode table rather than an assumption table.

Revisions made in the manuscript and Supplementary Information.

The changes appear in revised Tables 1 and 2. The table entries were shortened so that they work as compact reference tables rather than long prose blocks.

Relevant revised manuscript and supporting material.

Revised manuscript, Table 1.

Table 1 now reports descriptor name, symbol, unit or type, construction, and interpretation.

Revised manuscript, Table 2.

Table 2 now compares analytic response generation, surrogate prediction, feasible projection, physical stress test, and climate sensitivity by input, output, role, and boundary.

■ **Comment R1-7.**

The reviewer requested stronger justification of the methodology, validation studies using published or experimental data where possible, and clearer model assumptions.

Response.

✓ We agree that the assumptions and validation scope needed to be stated more clearly. We added the analytic response-generation equations and clipping ranges in the Supplementary Information, expanded the surrogate assessment, and added a 24-case physics-based cross-model stress test. The stress test completed

all locked cases, but the metric agreement and rank consistency are weak. We therefore present it as an interpretation boundary rather than as evidence that the analytic response surface is physically equivalent to EnergyPlus/Radiance outputs.

Revisions made in the manuscript and Supplementary Information.

The changes appear in Sections 2.2, 2.3, and 3.5, Table 6, Figures 9 and 10, and Supplementary Sections S1, S2, S5, and S6.

Relevant revised manuscript and supporting material.

Supplementary Information, Section S1.4.

The analytic EUI_t, EG, and H equations are now given explicitly, including density, openness, aspect-ratio, sky-view, orientation, open-space-position terms, stochastic terms, and clipping ranges.

Revised manuscript, Section 3.5 and Table 6.

The 24-case physics-based cross-model stress test is reported as complete, with weak agreement and unsupported ranking transfer. For the 18 direct feasible cases, EUI_t, the EG proxy, and H have nMAE values of 0.340, 0.405, and 0.601, with Spearman correlations of -0.404, -0.023, and -0.043. The EG stress-test output is stated as the GHI-based rooftop-PV proxy used in the locked physical case set.

■ **Comment R1-8.**

The reviewer requested an explanation of five-fold cross validation.

Response.

✓ We added a clearer explanation of the repeated five-fold protocol. The revised text states that the data are split into five folds, the model is trained on four folds and evaluated on the held-out fold, and the procedure is repeated with different fold assignments before reporting averaged metrics.

Revisions made in the manuscript and Supplementary Information.

The explanation appears in Section 2.3 and Supplementary Section S2.2. The corresponding quantitative summary appears in Table 3.

Relevant revised manuscript and supporting material.

Revised manuscript, Section 2.3 and Table 3.

Repeated five-fold validation gives mean nMAE values of 0.0176 for EUI_t, 0.0265 for EG, and 0.0145 for H, with corresponding Spearman correlations of 0.991, 0.984, and 0.995.

■ **Comment R1-9.**

The reviewer suggested clarifying the framework components.

Response.

✓ We reorganized the workflow description so that readers can distinguish fast surrogate-based search from post-optimization feasible-block assessment. The revised framework separates morphology generation, analytic target assignment, surrogate optimization, feasible projection, and selected sensitivity analyses.

Revisions made in the manuscript and Supplementary Information.

The change appears in Figure 1 and Section 2, with additional terminology support in Table 2.

Relevant revised manuscript and supporting material.

Revised manuscript, Section 2 and Table 2.

The workflow now distinguishes generated feasible block geometries, descriptor-space candidates, selected surrogate predictions, nearest-feasible-block projection, and locked physical and climate case calculations.

■ **Comment R1-10.**

The reviewer requested a reference for Markov Decision Process.

Response.

✓ We revised the RL framing instead of forcing the morphology problem into a natural MDP interpretation. The manuscript now presents DDPG as a continuous-action policy-learning comparator for serialized static black-box search on a surrogate response surface, while citing foundational RL and DDPG literature.

Revisions made in the manuscript and Supplementary Information.

The changes appear in Sections 1.2 and 2.4. The revised text cites foundational RL and DDPG studies while clarifying that the present environment is a surrogate-query formulation rather than physical-time control.

Relevant revised manuscript and supporting material.

Revised manuscript, Sections 1.2 and 2.4.

The revised text states that static urban morphology optimization differs from operational control because each query is an absolute candidate description evaluated by a surrogate response surface. DDPG is therefore used as a policy-learning comparator for generating descriptor queries in a bounded surrogate environment.

■ **Comment R1-11.**

The reviewer noted that Figure 3 lacked sufficient explanation and requested explanations of the symbols and equations.

Response.

✓ We revised the Figure 3 caption and the Methodology text to explain the actor, critic, target networks, replay buffer, TD target, critic loss, actor objective, and soft target-network update. The actor input and output are now tied to the normalized target state and absolute descriptor action.

Revisions made in the manuscript and Supplementary Information.

The change appears in Figure 3 and Section 2.4.

Relevant revised manuscript and supporting material.

Revised manuscript, Section 2.4 and Figure 3.

The revised DDPG subsection defines normalized target coordinates, the ideal vector, the weighted normalized distance, and the reward. Figure 3 summarizes the actor-critic and target-network structure used for the surrogate-query updates.

■ **Comment R1-12.**

The reviewer requested consistent parity-plot limits, matching axes, and units for each axis.

Response.

✓ We revised the parity plots and unit wording so that the x- and y-axes use matching limits and target units. The revised figures and table headings consistently use EUIt in $\text{kWh m}^{-2} \text{y}^{-1}$, EG in million kWh y^{-1} , and H in hours.

Revisions made in the manuscript and Supplementary Information.

The changes appear in Figures 4, 5, and 9, Table 6, and Supplementary Sections S2 and S5.

Relevant revised manuscript and supporting material.

Revised manuscript, Figures 4 and 9.

The surrogate parity and cross-model stress-test plots now use one-to-one references, target-specific units, and matching axis interpretation.

Supplementary Information, Sections S2 and S5.

The residual and physical stress-test summaries report target-specific units and make the analytic-target and physics-based quantities explicit.

■ **Comment R1-13.**

The reviewer requested an explanation of the meaning of episode in the training plots.

Response.

✓ We clarified that an episode is not a period of physical urban evolution. In the revised manuscript, each DDPG episode contains 40 surrogate queries, and the curves show policy-training behaviour on a static response surface.

Revisions made in the manuscript and Supplementary Information.

The change appears in Sections 2.4 and 3.2, and in the Figure 2 and Figure 6 captions.

Relevant revised manuscript and supporting material.

Revised manuscript, Section 2.4.

The revised text states that each episode begins from a random descriptor query and proceeds through absolute descriptor actions evaluated by the guarded surrogate. The 40-step horizon is a fixed sequence of policy-training queries, not physical time evolution.

■ **Comment R1-14.**

The reviewer suggested more generalized case studies, including different locations or building types.

Response.

✓ We added a limited cross-climate sensitivity analysis using Harbin, Beijing, and Guangzhou weather files for four selected generated blocks. We did not claim general cross-climate transfer and did not add new building-type families beyond the present residential prototype setting. The building-type extension is now stated as future work.

Revisions made in the manuscript and Supplementary Information.

The changes appear in Sections 2.6, 3.5, 4.2, and 4.3, Figure 10, Table 6, and Supplementary Section S6.

Relevant revised manuscript and supporting material.

Revised manuscript, Figure 10 and Table 6.

The climate analysis evaluates four selected blocks under Harbin, Beijing, and Guangzhou weather relative to the baseline weather. EG ordering remains stable in this four-block set, whereas H ordering and severe-cold EUIt ordering are climate-dependent.

Revised manuscript, Section 4.3.

The limitations now state that future work should incorporate additional building types, storage, EV

charging, distributed-energy interactions, and climate-aware surrogates across multiple regions.

■ **Comment R1-15.**

The reviewer suggested writing the paper for a broader audience.

Response.

✓ We revised the manuscript to improve readability for Applied Energy readers. The Introduction now starts from block-scale urban energy planning, the Abbreviations block supports the terminology, Table 2 separates evaluation modes, and the Discussion explains what can and cannot be inferred from descriptor-space, projection, physical, and climate analyses.

Revisions made in the manuscript and Supplementary Information.

The changes appear throughout the Introduction, front matter, Methodology tables, Discussion, and Supplementary Information.

Relevant revised manuscript and supporting material.

Revised manuscript, Sections 1 and 4.

The revised paper now separates the design motivation, optimizer-comparison question, representation issue, and sensitivity-analysis boundaries before presenting the technical results.

► Reviewer 2.

✓ We thank Reviewer 2 for the detailed methodological comments. These comments directly shaped the second-round revision. The revised manuscript now distinguishes analytic target assignment, surrogate prediction, feasible projection, selected physics-based case calculations, and cross-climate sensitivity, while reducing claims that are not supported by the current validation mode.

■ Comment R2-1.

The reviewer asked why the dataset was generated through an analytic approximation of the original EnergyPlus and Radiance workflow rather than through direct simulation.

Response.

✓ We agree that the original manuscript did not explain the data source with sufficient precision. We have revised the manuscript to state explicitly that the 2000-sample benchmark is an analytic response-generation dataset. The analytic generator provides a controlled demand-generation-sunlight response surface for comparing optimizer behaviour under a shared surrogate environment. This design makes repeated surrogate training checks and 20-seed optimizer comparisons feasible under 24,000 surrogate queries per seed or run, while keeping all optimizers on the same target-assignment surface. We now state that this dataset does not replace direct annual physical simulation for the complete generated morphology pool.

Revisions made in the manuscript and Supplementary Information.

The changes appear in Sections 2.1–2.3 and Supplementary Section S1.

Relevant revised manuscript and supporting material.

Revised manuscript, Section 2.2.

The analytic response generator is now described as the target-assignment module that maps a generated morphology descriptor vector to EUI_t, EG, and H. The text states that the physics-based case calculations are conducted subsequently on locked generated blocks.

Supplementary Information, Section S1.4.

The complete equations, stochastic terms, and clipping ranges are provided for EUI_t, EG, and H, including orientation, aspect-ratio, density, openness, sky-view, and open-space-position terms.

■ Comment R2-2.

The reviewer noted that the terms physical probe and physical-stack probe could imply stronger physical modelling than the study actually performs.

Response.

✓ We agree. We removed the stronger terminology and now use limited physics-based cross-model stress test. The revised text explains the completed EUIt path, the GHI-based rooftop-PV proxy used for EG, and the direct-sun-hour representation used for H. We also state that annual roof-plane irradiance output was unavailable in the locked case set, so the EG comparison is deliberately described as a proxy-based result.

Revisions made in the manuscript and Supplementary Information.

The changes appear in Table 2, Section 2.6, Section 3.5, Table 6, and Supplementary Sections S5.3 and S5.4.

Relevant revised manuscript and supporting material.**Revised manuscript, Table 2 and Section 3.5.**

The evaluation-mode table distinguishes physical stress test from analytic response generation and surrogate prediction. The Results section states that the EG stress-test output is the GHI-based rooftop-PV proxy used in the locked physical case set.

Supplementary Information, Section S5.4.

The Supplementary Information states that the locked physical workflow did not produce annual roof-plane irradiance outputs for the direct stress-test cases.

■ Comment R2-3.

The reviewer stated that the validation evidence remained insufficient and that the physical results showed noticeable deviations from surrogate predictions, especially for EUIt and H.

Response.

✓ We agree that the physical case results should not be presented as strong validation. We added the 24-case locked stress-test set, consisting of 18 direct feasible cases and 6 optimizer-linked cases, and revised the interpretation. The direct feasible cases show weak metric agreement and unsupported ranking transfer: EUIt nMAE is 0.340 with Spearman $\rho = -0.404$, the EG GHI proxy has nMAE = 0.405 with Spearman $\rho = -0.023$, and H has nMAE = 0.601 with Spearman $\rho = -0.043$. The manuscript now treats this result as an interpretation boundary rather than as confirmation of optimizer ranking.

Revisions made in the manuscript and Supplementary Information.

The changes appear in Sections 2.6, 3.5, and 4.2, Figure 9, Table 6, and Supplementary Section S5.

Relevant revised manuscript and supporting material.**Revised manuscript, Section 3.5 and Table 6.**

The revised manuscript reports that the cross-model stress test completed all locked cases but that

Spearman correlations are negative or near zero for EUI_t, the EG proxy, and H in the 18 direct feasible cases.

Supplementary Information, Section S5.3.

The per-case physical summary separates the 18 direct feasible cases from the 6 optimizer-linked cases because the latter include an additional projection step from continuous descriptor-space candidates to nearest generated feasible blocks.

■ **Comment R2-4.**

The reviewer requested the sample-generation strategy and distributional coverage of the 12-dimensional design space.

Response.

✓ We expanded the sampling description. The revised manuscript states that the generated block is a 240 m by 240 m site divided into a three-by-three grid; the generator samples open-space position, building prototype, floor count, and orientation before calculating the 12 morphology descriptors. Coverage is now summarized through descriptor distributions, OSLI counts, nearest-neighbour distances, and PCA structure. We also added dependency checks showing that FAR is linked to BD and AF, OSR is linked to footprint and gross floor area, OSLI is integer-valued, exact duplicate descriptor rows are absent, and six principal components are needed to explain 95% of descriptor variance.

Revisions made in the manuscript and Supplementary Information.

The changes appear in Section 2.1, Figure 4, Table 1, and Supplementary Sections S1.2, S1.3, and S1.5.

Relevant revised manuscript and supporting material.

Supplementary Information, Sections S1.2–S1.5.

The Supplementary Information reports the random morphology-generation sequence, confirms integer OSLI consistency, reports no exact duplicate descriptor rows in the 2000-row pool, gives the FAR minus BD times AF residual as 1.33×10^{-15} , and reports that six principal components explain 95% of descriptor variance.

■ **Comment R2-5.**

The reviewer questioned the RL formulation because the problem is a static black-box optimization problem rather than a naturally sequential decision-making process.

Response.

✓ We agree with this concern and have rewritten the DDPG formulation accordingly. The revised manuscript states that DDPG is used as serialized static black-box search on a surrogate response surface. The state is the normalized target output, the action is an absolute 12-descriptor vector, and one episode is a fixed sequence of 40 surrogate queries. The episode does not represent physical time evolution.

Revisions made in the manuscript and Supplementary Information.

The changes appear in Sections 1.2, 2.4, and 3.2, and in Figures 2, 3, and 6.

Relevant revised manuscript and supporting material.

Revised manuscript, Section 2.4.

At each step, the actor outputs an absolute 12-dimensional normalized descriptor action. This action is not an incremental modification of the preceding block. The 40-step horizon functions as a fixed sequence of policy-training queries on a static surrogate response surface.

■ **Comment R2-6.**

The reviewer requested clearer distinction among the analytic response-generation workflow, surrogate-based evaluation, selected physical case checks, and stronger physical-evaluation terminology.

Response.

✓ We removed or replaced the confusing stronger terms and added a compact evaluation-mode table. The revised terminology is analytic response generation, surrogate prediction, feasible projection, physics-based cross-model stress test, and cross-climate sensitivity. The manuscript no longer presents the selected case calculations as a strong physical ranking result, and the response text consistently separates analytic-target accuracy from selected physics-based stress testing.

Revisions made in the manuscript and Supplementary Information.

The changes appear in Table 2 and throughout Sections 2.2, 2.6, 3.5, and 4.2.

Relevant revised manuscript and supporting material.

Revised manuscript, Table 2.

The evaluation-mode table now reports each mode by input, output, role, and boundary, separating target assignment, optimizer-time surrogate prediction, nearest-neighbour feasible projection, selected cross-model case calculations, and weather-file sensitivity.

■ **Comment R2-7.**

The reviewer requested clear formulas or workflows for calculating EUI_t , EG , and H in the generated dataset.

Response.

✓ We added the analytic response equations in the Supplementary Information and summarized their role in the main manuscript. The response directions are now explicit: lower EUIt is preferred, while higher EG and H are preferred. The SI equations define the orientation penalty, aspect-ratio mean, morphology sweet-spot term, density penalty, stochastic terms, and clipping ranges used by the analytic target-assignment surface.

Revisions made in the manuscript and Supplementary Information.

The changes appear in Section 2.2 and Supplementary Section S1.4.

Relevant revised manuscript and supporting material.**Supplementary Information, Section S1.4.**

The Supplementary Information defines the analytic EUIt, EG, and H responses as functions of morphology descriptors, orientation, open-space position, derived openness terms, stochastic terms, and clipping ranges; the main manuscript keeps the equations summarized to avoid overloading the Methodology section.

■ Comment R2-8.

The reviewer requested code, data, or detailed reproducibility information.

Response.

✓ We revised the Data and code availability statement. The public release will provide the code, processed datasets, result tables, and reproducibility notes required to regenerate the analytic response dataset, surrogate assessment summaries, optimizer comparison tables, feasible-projection results, physics-based stress-test summaries, and cross-climate sensitivity results. Local credentials and private raw artifacts are excluded.

Revisions made in the manuscript and Supplementary Information.

The change appears in the Data and code availability section.

Relevant revised manuscript and supporting material.**Revised manuscript, Data and code availability.**

The revised statement gives the public repository as <https://github.com/haolinwang1313/DRL-Driven-Optimization> and identifies the processed datasets, scripts, result tables, and reproducibility notes planned for release.

► Reviewer 3.

✓ We thank Reviewer 3 for identifying important consistency issues in the workflow figures, utility interpretation, reward equation, units, and wording about simulation and physical probing. These comments helped align the figures, equations, and claims with the actual revision evidence.

■ Comment R3-1.

The reviewer requested correction of spelling and wording errors in the workflow figures.

Response.

✓ We corrected the workflow and DDPG-architecture figures. The visible errors noted by the reviewer were removed, and the revised figures now use labels such as locations, Actor network, and Critic network.

Revisions made in the manuscript and Supplementary Information.

The changes appear in Figures 1 and 3 and their captions.

Relevant revised manuscript and supporting material.

Revised manuscript, Figures 1 and 3.

The revised figures were replaced by final manual vector-PDF candidates and checked for the reviewer-noted spelling problems before manuscript integration.

■ Comment R3-2.

The reviewer requested checking the consistency between the utility panel and the benchmark conclusion.

Response.

✓ We reworked the optimizer-comparison narrative and figure structure. The revised benchmark no longer states a simple NSGA-II superiority claim across all quantities. It separates fixed-domain post-hoc utility, equal-size descriptor-space HV and IGD, output contracts, full-archive interpretation, and feasible-block projection.

Revisions made in the manuscript and Supplementary Information.

The changes appear in Section 3.3, Table 5, Figure 7, Figure 8, and Supplementary Section S3.

Relevant revised manuscript and supporting material.

Revised manuscript, Section 3.3 and Table 5.

The equal-size descriptor-space comparison reports NSGA-II with $HV = 1.330995$ and $IGD = 0.004710$ at retained size 20. DDPG remains scenario-dependent, with HV values of 0.661912, 1.002757, and 1.170428 across the three preference settings.

Revised manuscript, Section 3.4.

The projection analysis reports that the 2000-candidate NSGA-II descriptor archive maps to 51 unique generated feasible blocks, so descriptor-space archive size is not interpreted as feasible-morphology diversity.

■ **Comment R3-3.**

The reviewer requested rechecking the mathematical form of the weighted-distance reward.

Response.

✓ We corrected the reward formulation. The revised manuscript defines normalized target coordinates, the ideal vector $\mathbf{u} = [0, 1, 1]^T$, the weighted normalized distance $d_w = \|\mathbf{w} \odot (\mathbf{z} - \mathbf{u})\|_2 / \|\mathbf{w}\|_2$, and the reward $R = 1 - d_w$. We also clarify that $\mathbf{w}$ acts as an axis-scaling coefficient in normalized target space and that this training reward is distinct from the fixed-domain post-hoc utility used for benchmark comparison.

Revisions made in the manuscript and Supplementary Information.

The correction appears in Section 2.4 and Figure 3.

Relevant revised manuscript and supporting material.

Revised manuscript, Section 2.4.

$$d_w = \frac{\|\mathbf{w} \odot (\mathbf{z} - \mathbf{u})\|_2}{\|\mathbf{w}\|_2}, \quad R = 1 - d_w.$$

The revised text also states that Balanced uses $\mathbf{w} = [1/3, 1/3, 1/3]^T$, Energy Saving uses $\mathbf{w} = [0.6, 0.2, 0.2]^T$, and Energy Generation uses $\mathbf{w} = [0.2, 0.6, 0.2]^T$.

■ **Comment R3-4.**

The reviewer requested standardization of units in figure axes and table headings.

Response.

✓ We standardized units across the revised figures and tables. EUI is reported in $\text{kWh m}^{-2} \text{y}^{-1}$, EG is reported in million kWh y^{-1} , H is reported in hours, theta is reported in degrees, and SD is reported in metres where applicable.

Revisions made in the manuscript and Supplementary Information.

The changes appear in Tables 1, 3, 5, and 6, Figures 4–10, and the Supplementary Information tables and captions.

Relevant revised manuscript and supporting material.

Revised manuscript and Supplementary Information, tables and figure captions.

The revised package uses consistent units for target values, residuals, morphology descriptors, projection distances, and climate deltas.

■ Comment R3-5.

The reviewer requested consistent wording about simulation, analytic approximation, and physical probing.

Response.

✓ We revised this wording throughout the manuscript. The benchmark dataset is now described as an analytic response-generation dataset, the 2000 generated samples are not described as direct physical simulations, and the selected case calculations are described as a limited physics-based cross-model stress test.

Revisions made in the manuscript and Supplementary Information.

The changes appear in the Abstract, Sections 1–4, Table 2, Table 6, and the Supplementary Information.

Relevant revised manuscript and supporting material.

Revised manuscript, Abstract and Sections 2.2 and 4.2.

The revised text states that the optimizer comparison is conducted on an analytic response-generation dataset, while direct physical calculations are available only for locked stress-test cases. The Discussion states that the sensitivity analyses do not convert the descriptor-space comparison into a physical ranking result.

► **Reviewer 4.**

✓ We thank Reviewer 4 for the comments on study necessity, climate scope, energy-system extensions, best-scenario definition, and comparison with existing urban energy planning approaches. The revised manuscript strengthens the motivation while keeping the present evidence within the block-morphology, rooftop-generation, sunlight-access, and limited sensitivity-analysis scope.

■ **Comment R4-1.**

The reviewer asked that the necessity of the study be highlighted more clearly in relation to existing block-scale urban energy planning studies.

Response.

✓ We revised the Introduction to clarify that the study is not simply another optimizer application. Its purpose is to show how optimizer ranking in early-stage block-scale urban energy design depends on surrogate conditioning, descriptor representation, retained-output structure, and feasible-block projection.

Revisions made in the manuscript and Supplementary Information.

The changes appear in Sections 1.1–1.3 and the Discussion.

Relevant revised manuscript and supporting material.

Revised manuscript, Sections 1.1–1.3.

The revised Introduction connects block morphology to building demand, rooftop solar access, and winter sunlight availability, and then identifies the methodological gap: policy learning and Pareto search need careful comparison when both operate on a shared surrogate response surface and continuous descriptor outputs must be mapped back to feasible morphologies.

■ **Comment R4-2.**

The reviewer asked how the selected climate condition affects the modelling and how the findings could be translated into other climate zones.

Response.

✓ We added a limited cross-climate sensitivity analysis using Harbin, Beijing, and Guangzhou weather files for the same four selected generated blocks. The results are presented as sensitivity results, not as a general transferability claim. Mean EUI changes are +55.681 for Harbin, +13.467 for Beijing, and -2.972 for Guangzhou relative to the baseline weather. EG ordering remains stable in the four-block set, while H ordering is climate-dependent and Harbin changes the EUI ordering.

Revisions made in the manuscript and Supplementary Information.

The changes appear in Sections 2.6, 3.5, and 4.2, Figure 10, Table 6, and Supplementary Section S6.

Relevant revised manuscript and supporting material.

Revised manuscript, Figure 10 and Table 6.

The cross-climate analysis evaluates four blocks under Harbin, Beijing, and Guangzhou weather relative to the baseline weather. Supplementary Section S6 reports EG Spearman $\rho = 1.0$ in all three additional climates, H $\rho = 0.2, 0.4$, and 0.2 for Beijing, Guangzhou, and Harbin, and Harbin EUIt $\rho = 0.2$. The revised text states that this is a limited sensitivity analysis and that the surrogate is not retrained for each climate.

■ Comment R4-3.

The reviewer asked how different energy sources, the EV market, and distributed energy systems may affect urban energy planning.

Response.

✓ We clarified the present objective scope and added this topic to the limitations and future-work discussion. The current study considers demand, rooftop-generation proxy, and winter sunlight hours. It does not model storage, EV charging, district-level grid exchange, or distributed-energy operational interactions.

Revisions made in the manuscript and Supplementary Information.

The change appears in Section 4.3.

Relevant revised manuscript and supporting material.

Revised manuscript, Section 4.3.

The limitations now state that future work should incorporate storage, EV charging, and distributed-energy interactions in addition to expanding the physical simulation set and training climate-aware surrogates across multiple regions.

■ Comment R4-4.

The reviewer noted that the definition of the best scenario in the optimization process was unclear.

Response.

✓ We revised the optimizer-comparison protocol to avoid ambiguous best-scenario wording. The manuscript now distinguishes best scalarized DDPG candidate, fixed-domain post-hoc utility, descriptor-space HV and IGD, output-contract size, projected feasible blocks, and selected physical case results.

Revisions made in the manuscript and Supplementary Information.

The changes appear in Sections 2.5, 3.3, and 3.4, Tables 4 and 5, and Figures 7 and 8.

Relevant revised manuscript and supporting material.

Revised manuscript, Tables 4 and 5.

Table 4 defines optimizer budgets and retained outputs. Table 5 reports the descriptor-space comparison using a fixed normalized reference, so each performance claim is tied to a stated metric and retained-output contract.

■ **Comment R4-5.**

The reviewer requested more comparison with existing urban energy planning frameworks to validate the advantages of the proposed framework.

Response.

✓ We expanded the literature review and Discussion to position the study against conventional surrogate optimization, NSGA-II Pareto archives, DRL operational-control applications, and derivative-free energy-system search. The revised manuscript does not claim comprehensive superiority over all existing approaches. It states the practical advantage more narrowly: the workflow separates descriptor-space search, feasible-block projection, and selected sensitivity analysis so that optimizer comparisons can be interpreted more transparently.

Revisions made in the manuscript and Supplementary Information.

The changes appear in Sections 1.2, 4.1, and 4.3.

Relevant revised manuscript and supporting material.

Revised manuscript, Sections 1.2 and 4.1.

The revised literature review discusses parametric design, evolutionary multi-objective optimization, surrogate modelling, DRL in operational energy control, and neighbouring derivative-free methods. The Discussion states that the method comparison is strongest as a representation-aware design-support workflow, not as a universal ranking of optimizers.

Closing

We again thank the editor and reviewers for the detailed comments. The revised manuscript and Supplementary Information have been reorganized to make the data-generation process, optimizer formulation, comparison protocol, physical and climate scope, and remaining limitations clearer. We hope that the revised package addresses the reviewers' concerns and provides a more transparent basis for evaluating surrogate-conditioned optimizer comparisons in block-scale urban energy design.